

Title

CAMPUS NAVIGATOR ASSISTANT

A Python-based conversational and map-guided system to help new visitors and students navigate the university

Abstract

New students and visitors often struggle to locate departments, classrooms, offices, events, and services on their first visits to campus. The Campus Navigator Assistant combines a conversational chatbot with map-based navigation to provide step-by-step directions, place information, and FAQs. Built with a Python backend (FastAPI/Flask), an NLP layer (Rasa/OpenAI/Transformers), a geospatial store, and a responsive UI (web and/or desktop/mobile wrapper), the system allows users to ask natural questions (e.g., “How do I get from the main gate to the CS department?”) and receive guidance through indoor/outdoor maps, shortest-path routing, and contextual answers. The project delivers a working prototype with a clean architecture, database schema, annotated datasets, and documentation within four weeks.

Introduction

Universities typically have complex layouts with multiple gates, departments, hostels, laboratories, libraries, and student service centers. Orientation signage may be insufficient during peak admission days, and printed maps quickly become outdated. A digital assistant that understands natural language queries and offers reliable routing reduces confusion, improves campus experience, and frees staff from repeated directional requests. This project proposes a Python-powered assistant that provides both chat and map interactions.

Problem Statement

Newcomers lack a single, up-to-date, and interactive source of truth for campus navigation and basic information. Existing resources (static PDFs, disparate webpages, ad-hoc directions) are fragmented and do not support turn-by-turn guidance or natural questions. The absence of centralized FAQs and real-time routing leads to lost time, missed appointments, and avoidable congestion at help desks.

Objectives

1. Provide accurate indoor/outdoor navigation across campus locations.
2. Enable natural-language Q&A for places, timings, contacts, and procedures.
3. Offer shortest-path routing with accessible route options (ramps/elevators).
4. Support landmark-based guidance and multi-stop directions.
5. Maintain an admin interface to update locations, events, and FAQs.
6. Log anonymized queries to improve coverage and model performance.
7. Deliver a secure, maintainable Python backend and a responsive UI.

Scope (prototype):

Campus map ingestion (geojson) and core POIs: gates, departments, offices, library, cafeteria, hostels, clinic, auditoriums, common labs.

Outdoor pathfinding; optional basic indoor routing for 1–2 buildings (floor graph).

Web UI with chat + map, English UI copy; mobile-responsive layout.

Admin panel (basic CRUD) for POIs, categories, and FAQs.

Scope

Real-time crowd density or live bus tracking.

Full multilingual support (can be future work).

Complete indoor mapping of all buildings.

Literature Review (brief)

Campus and venue navigation assistants rely on graph-based pathfinding (Dijkstra/A*), with POIs stored in spatial databases (PostGIS) or graph stores.

Conversational campus bots show improved orientation outcomes when combined with visual maps and contextual cards.

Industry practice: use of Google Maps Platform for base layers, Directions, Places, and Geocoding; or open-source stacks (OpenStreetMap + Leaflet/MapLibre + OSRM/pgRouting).

Privacy-preserving analytics and rate-limiting are recommended for public kiosks and web apps.

Tools & Technology

Programming Language: Python

Backend Framework: FastAPI (preferred) or Flask

*Routing: NetworkX/pgRouting for on-campus graph; optional Google Maps
Directions API integration*